Construction Technology of the Curved 'Mars House' Digital Hotel Project in China Using Concrete 3D Printing

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Abstract. 3D printing technology is one of the most likely construction technologies to be used when building extraterrestrial bases in the future. The Mars 1 Base project in Gansu Province, China simulates the Martian landscape and uses concrete 3D printing technology to construct houses, creating a unique cultural tourism industry project. In June 2022, the 3D printing construction of five Martian hotel buildings began. The 3D-printed hotel is egg-shaped, with a total height of 9 m and a floor area of 46 square meters, divided into two floors with triangular windows and a skylight on the top. After the concrete components are 3D printed in the factory, they are transported to the project site for assembly and construction. This article introduces the structural design, 3D printing process and overall installation and construction of the Mars house. The five 3D-printed Martian houses and hotels were built in the Gobi Desert and took three months to complete. Based on digital printing and concrete 3D printing construction, the entire printing, installation, and construction process was safe, fast, and accurate, with a complete structural design and installation process. This project is currently the world's largest egg-shaped 3D printed structure, and its construction process provides experience for more 3D printing designs and is of positive significance for exploring the addition of intelligent building technology.

Keywords: Concrete 3D printing · Mars house · structural design · construction process · 3D printing technology

1 Introduction

Since the latter half of the 20th century, significant progress has been made in Mars research, allowing scientists to gain a better understanding of the planet's geology, climate, and history (Longhi et al. 1992; Burr et al. 2002; Cousins et al. 2011). At the same time, the advancement of Mars exploration and space missions has sparked interest in Mars colonization and habitat development (Muscatello et al. 2012; Ho et al. 2022; Rüede et al. 2019). These efforts aim to improve the feasibility of Mars exploration missions and ultimately enable humans to inhabit Mars long-term (Sen et al. 2010; Naser 2019).

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Concrete 3D printing technology presents a promising approach to constructing Martian habitats. This technology has already made significant strides in the construction industry on Earth (Khoshnevis 2004; Kazemian et al. 2022; Nematollahi et al. 2017; Ma et al. 2022; Schutter et al. 2018) and has sparked discussions about its potential application in Martian habitat construction. Building Martian habitats presents numerous challenges, such as extreme environments, limited resources, and the incompatibility of traditional building materials and techniques (Reches 2019). These challenges drive researchers and engineers to seek innovative building methods and in-situ resource utilization strategies.

Currently, various countries are showcasing their plans and progress in 3D printing Martian habitats. For example, NASA has organized the 3D-Printed Habitat Challenge to inspire innovation in Martian habitat design (Muthumanickam et al. 2020; Kading et al. 2015). Meanwhile, the European Space Agency (ESA) is investigating the feasibility of 3D printing technology for constructing lunar habitats in its Moon Village project. This endeavour aims to create a sustainable lunar settlement that could serve as a stepping stone for Mars exploration (Blau 2016). Furthermore, the Chinese government has announced a future five-year space exploration plan, including the joint construction of an international lunar science research station and a Mars sample return mission (China Government Website 2022).

In China, the city of Jinchang is promoting a simulated Martian tourism project called the "Mars-1 Base" (China.com.cn 2022). The "Mars Habitat Digital Hotel," consisting of five houses built using concrete 3D printing technology, is an important part of the project. The Zhejiang University Intelligent Construction Team will use advanced concrete 3D printing technology to construct these houses layer by layer, significantly reducing construction costs and time. The purpose of this article is to introduce the construction process and technological implementation of the Mars Habitat Digital Hotel project, including architectural structure design, concrete 3D printing process, and overall installation and construction technology.

2 Architectural Design and Structural Design of Mars House

The egg-shaped structure is relatively easy to achieve in 3D printing technology, and this design can better withstand the harsh conditions of the Martian environment, such as strong winds, radiation, and high-temperature differences. According to the characteristics of the project construction site's environment, terrain, and the purpose of providing reliable accommodation, the owner chose to design an egg-shaped building form and use concrete 3D printing technology for rapid construction. This technology is faster, cheaper, and more durable than plastic printing construction. The entire Mars egg-shaped house is 9 m high with a floor area of 46 square meters, divided into two floors with triangular window design, and a skylight on the top for viewing. As shown in Fig. 1.

Fig. 1. The architectural design and segmented 3D printing model of the Mars house.

The Mars house's overall structure utilizes a concrete 3D printed wall as the outer shell, with a steel mesh arranged inside the printing mould. After printing, concrete is poured inside the mould to form a reinforced concrete structure, ensuring the stability of the 3D-printed permanent mould and the safety of the entire building. The structure, which stands 9 m high, is divided into four rings for printing, with structural columns set in each ring. As shown in Fig. 2.

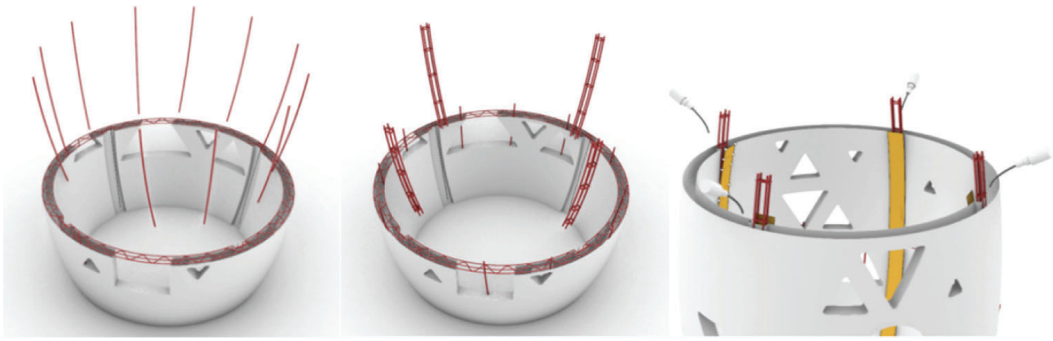

Fig. 2. The structural design scheme of the Mars house.

The structural design and reinforcement method for the Mars house considered factors such as horizontal reinforcement, vertical reinforcement, and structural columns. During the printing process of each ring-shaped structure, the circumferential horizontal reinforcement mesh is first placed, and the reinforcement between the steel bars is welded into a whole. These reinforcement meshes not only reinforce the printing structure but also serve as hoop reinforcement for the overall structure. A layer is set every 25 cm to enhance the structural strength and interlocking of the 3D-printed outer shell and to improve the overall performance and stability of the structure. After each outer shell is printed, a circle of φ12 mm vertical reinforcement needs to be inserted inside and tied with horizontal reinforcement as much as possible to enhance the bearing capacity and stability of the entire structure. Then, C40 self-compacting concrete is poured to form a composite reinforced concrete structure of the outer shell and the inner core concrete. The foundation structure of the Mars house adopts a traditional reinforced concrete structure.

In summary, the architectural design and reinforcement of the Mars house need to consider the special characteristics of the usage environment and building shape to ensure the safety and stability of the structure. The reasonable design of steel reinforcement can optimize the load-bearing performance of the concrete structure, thus ensuring the durability and safety of the Mars house. The use of advanced 3D printing technology and automation control systems can improve the efficiency and quality of construction.

3 3D Printing Construction Technology and Process

3.1 Mars House 3D Printing Process

The 3D printing process for Mars houses includes starting from the model refinement design, preparing personnel, materials, and equipment, followed by printing construction at the component factory, and then pouring and embedding lifting points for the components. Subsequently, the wall components are hoisted and transported to the site for on-site assembly, and finally, the house decoration construction is completed and delivered for use, as shown in Fig. 3.

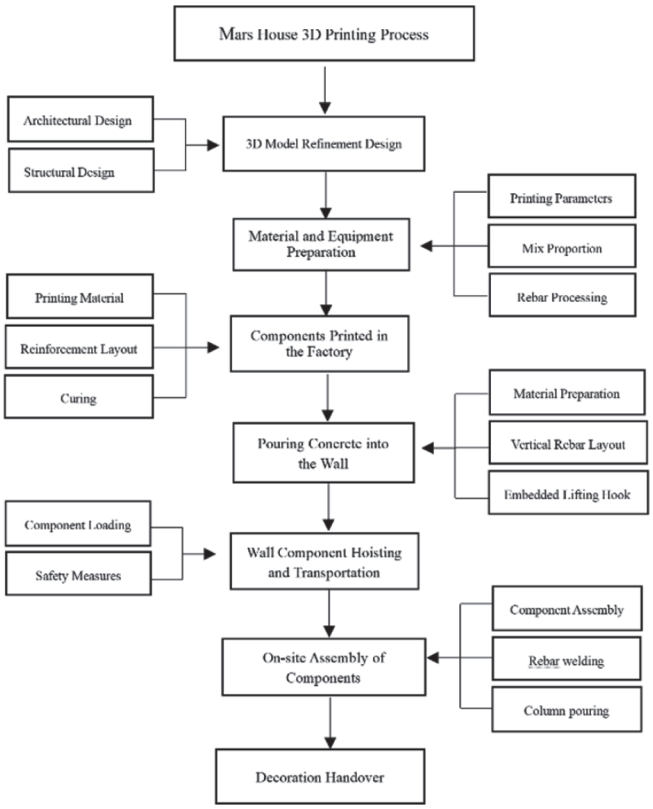

Fig. 3. The 3D printing process for Mars houses.

3.2 Personnel, Materials, and Machines for 3D Printing

The printing of house components uses 2 industrial-grade concrete 3D printers, as shown in Fig. 4., with effective printing dimensions of 14.8 m × 8.6 m × 5.5 m. The printing

process parameters are set to a single layer print height of 15 mm, print speed of 120 mm/s, and extrusion width of 40 mm for printing materials, only printing the outer frame of the house 3D model without setting internal filling paths. After importing the 3D model into the concrete 3D printer control software, the software automatically calculates the total amount of 3D printing materials for a single Mars house based on the set printing parameters, which is 10.6 m^3, with 17.76 m^3 of internal filling concrete, totalling 28.36 m^3.

The required reinforcement mainly includes ribbed steel bars with diameters of 10 mm and 12 mm, used for making steel mesh sheets and steel cages, with a total consumption of about 5 tons. In addition, auxiliary equipment such as cranes and transport trucks is required. To ensure smooth construction, corresponding personnel allocation and training are also necessary, including printer operators, construction workers, etc. The transportation distance from the printing factory to the project site is 1800 kms, and long-distance transportation also involves measures for component installation and transportation. The 3D printing and installation work of the 5 Mars houses were completed from June 25, 2022, to September 25, 2022, with an overall duration of 3 months.

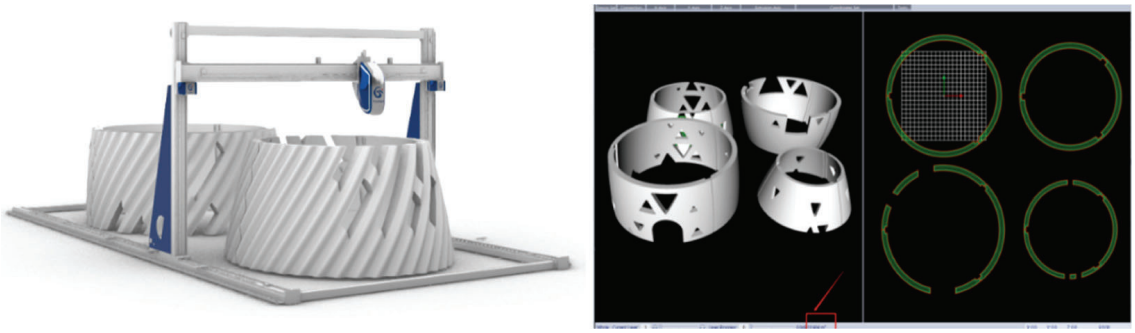

Fig. 4. Concrete 3D printer and control software

3.3 3D Printed Concrete Mix Proportion and Performance

The 3D-printed concrete used in this project has undergone strict experimental testing and meets various technical requirements. The concrete mix proportion is as follows in Table 1:

Table 1. The concrete mix ratio for 3D printing of the Mars house

Raw materials	OPC42.5 cement/%	Artificial sand/%	Admixture/%	Water/ %
Mix proportion	38	42	4	16

According to the testing methods specified in the Chinese standard "Technical Code for Concrete 3D Printing Technology" (T/CECS 786-2020), the tested concrete performs well in multiple performance indicators. Specifically, it exhibits a workability of 200 mm

and a setting time of 41 min, with good extrudability and supportability. After 28 days of curing, the concrete's mechanical performance was verified, the compressive strength of the test specimens made using standard moulding is 47.9MPa, while the compressive strength of the printed samples is 40.3MPa, with a strength reduction of 16%. Additionally, the concrete exhibits an interlayer tensile strength of 3.2MPa, interlayer tensile strength ratio of 128%, interlayer bonding strength of 2.7MPa, and interlayer bonding strength ratio of 92%. The interlayer tensile strength and interlayer bonding strength for the interface between old and new concrete are 3.2MPa and 2.5MPa, respectively. Furthermore, according to the test results of the Chinese standard GB/T 50082-2009, the 3D-printed concrete exhibits good resistance to permeation, with a permeability grade of P12. Moreover, the results of contact method testing at 28 days and 56 days show a stable and small amount of shrinkage, with values of 205×10^{-6} and 258×10^{-6}, respectively, indicating good volume stability. In summary, the 3D printing concrete performs well in terms of workability, mechanical performance, resistance to permeation, and volume stability, meeting the performance requirements for 3D printed construction.

3.4 3D Printing Prefabrication of Components

When prefabricating components for 3D printed houses in the factory, the 3D model of the building needs to be imported into the control software of the 3D printer first, and the basic parameters for printing, such as printing width and single-layer printing height, need to be set. According to the proportion of materials, cement, sand, additives, and water are weighed and mixed, and then pumped for printing. Each industrial-grade printer can print 2 ring-shaped structures at the same time, and 2 printers can simultaneously print 4 ring-shaped structures of an egg-shaped house. It takes 6 days to complete the printing of these 4 ring-shaped structures, followed by curing and pouring. Figure 5 shows four modules of Mars House printed by two different 3D printers.

Fig. 5. The 4 annular components of a Mars house have been 3D printed.

During the printing process, the prefabricated horizontal reinforcement mesh is manually placed on the printing path to enhance the mechanical performance of the printed wall. After the complete printing of the mould, vertical reinforcement and lifting rings are inserted, and C40 self-compacting concrete is poured. Before the start of printing each day, a 3D printing interface agent is applied between the new and old concrete interface layers to ensure good interlayer bonding strength between the cold joint layers.

After 3 days of curing, the printed ring-shaped structures will be cut and separated into smaller components at the preset column positions, and then stored in the storage yard. They will be transported to the project site for reassembly of the ring-shaped structure, with horizontal reinforcement and vertical reinforcement welded at the location of the construction column, and then sealed with mortar to form a structural column, as shown in Figs. 6. The entire process needs to be operated according to the printing process flow to ensure that the printed nest house components have excellent quality and meet the design requirements.

Fig. 6. The printing process of the components and the arrangement of the reinforcing bars.

Compared with on-site printing, printing components in the factory has the advantages of process controllability and mature site conditions, which is beneficial for the quality control of building components. The 3D printing process is a comprehensive new technology that combines multiple specialities and occupations such as architectural design, structure, materials, and printing implementation, and it has unique advantages in the structural formation of such irregular-shaped houses. It can significantly save the cost of template production and greatly reduce construction time.

4 Mar House On-Site Assembly and Construction

The completed printed wall components are transported by truck to the installation site and placed in the corresponding installation area. After levelling the installation site, a crane is used to assemble the wall components into four ring-shaped structures in a certain order, as shown in Fig. 7. Depicting the lifting process of the Mars habitat house.

Fig. 7. The transportation and assembly process of the annular components.

The reinforcement cages at the structural column positions of the assembled ring-shaped structures need to be welded, and mortar needs to be poured in the middle of the structural columns, leaving a 0.5-m reserved space at each end for connecting the columns after the installation of the whole structure, as shown in Figs. 8.

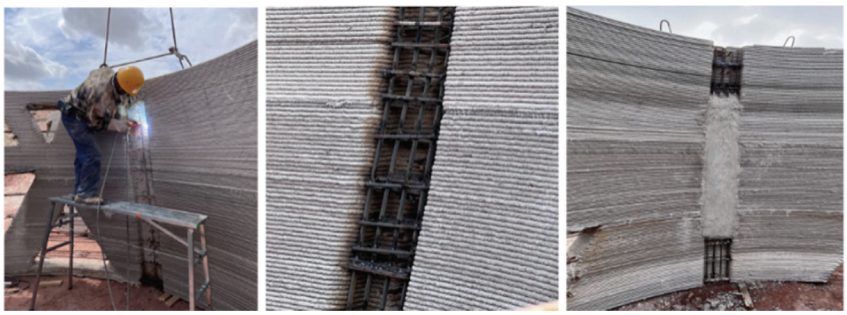

Fig. 8. Welding of reinforcement cages and concrete pouring for the components' columns.

Before the lifted ring-shaped structures are installed on the foundation base, the reinforced concrete foundation needs to be backfilled and levelled to ensure the verticality of the installed house. During the lifting process, a level is used to measure the installation of each ring-shaped structure, and levelling materials and measures are used to ensure the horizontal installation of each ring structure. The lifting process of the ring-shaped components is shown in Fig. 9.

Fig. 9. The foundation and assembly process of Mars house components.

In this project, the printed wall components were assembled one by one using a large crane, forming a huge ring-shaped structure. Over 10 sets of steel cables were used to lift the ring-shaped structure, and the lifting process was smooth and safe. The lifting and installation of the four ring-shaped structures took one day to complete. During the assembly process, the construction personnel used a level to measure the horizontalness of each wall component to ensure the verticality and stability of the house. Additionally, as shown in Fig. 10, after the completion of the house assembly, reinforcement welding of steel bars was performed at the reserved positions of the structural columns, and mortar was poured to form complete structural columns, strengthening the structural stability of the house.

Fig. 10. The connection process of the vertical structural columns inside the Mars house.

The entire installation and construction process of this habitat hotel house was safe and fast, thanks to the accuracy of the 3D printing digital printing construction process, as well as the relatively complete structural design and installation process. With the completion of the main body of the house, the subsequent interior and exterior wall decoration and construction can begin, as shown in Fig. 11.

Fig. 11. The interior and exterior walls of five installed Mars houses have been sprayed.

5 Summary

With the increasing trend of research and application of 3D printing technology in the field of construction, the successful application of concrete 3D printing in the Mars habitat hotel project has greatly improved the construction speed and shortened the construction period through the factory printing technology, where the wall components can be accurately printed and assembled on site. This unique project is the largest egg-shaped 3D printed structure domestically and internationally, and its construction process provides some technical experience for more 3D printed structures, which has positive significance for promoting the development of additive intelligent construction. This new construction technology will greatly change the development mode of the traditional construction industry, and bring more opportunities and challenges to the construction industry.

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